

Features:

1. Comes with ESP32-32E module, easy to develop, and sufficient development resources
2. 2.8-inch color screen, 240x320 resolution, maximum support for 262K colors (RGB666), rich display colors
3. Rich interfaces, convenient for connecting various peripherals (SPI, UART, etc.)
4. Support external speakers to play audio
5. Comes with RGB three color indicator lights, providing rich indication status
6. Comes with a resistive touch screen for convenient human-computer interaction
7. Standard TYPE-C interface, convenient for program download and power supply
8. Comes with a micro TF card slot for easy expansion of storage
9. Supports external lithium batteries, lightweight and portable
10. Comes with a battery charging management circuit to ensure safe charging and discharging of the battery
11. Provide rich example programs for easy learning
12. Provide technical support for underlying drivers and update WIKI materials online

Parameter:

Power supply interface: Type-C

Specifications:

Equipped with touch screen: E32R28T

No touch screen: E32N28T

Main control parameters:

Module: ESP32-32E

CPU: ESP32-D0WD-V3, Xtensa dual core 32-bit LX6 microprocessor

Main frequency: 240MHz (maximum)

Storage: 448KB ROM+520KB SRAM+16KB RTC SRAM+4MB external QSPI Flash

WIFI: 2.4GHz, 802.11b/g/n mode

Bluetooth: Bluetooth V4.2 BR/EDR and Bluetooth LE standards

Working voltage: 3.0~3.6 (V)

LCD screen parameters

Screen size: 2.8 inches

Screen type: TN TFT

Resolution: 240xRGBx320 (pixels)

Effective display area: 43.20 (W) x 57.60 (H) (mm)

Number of colors: Maximum: 262K (RGB666)

Commonly used: 65K (RGB565)

Driver IC: ST789P3

Display interface: 4-Line SPI (connected to ESP32)

Pixel size: 0.153 (H) x 0.153 (mm)

Viewing angle: 120 °CLOCK

Backlight brightness (typical value): 260 cd/m²

Backlight type: White LED * 4

Working temperature: -10~50 (°C)

Storage temperature: -20~60 (°C)

Touch screen parameters

Effective area size: 2.8 inches

Touch screen type: resistive touch screen

Driver IC: XPT2046

Effective touch area: 44.20 (W) x 58.60 (H) (mm)

Visual window size: 45.20 (W) x 59.45 (H) (mm)

Communication interface: SPI

Structural material: ITO film+ITO glass

Working temperature: -10~60 (°C)

Storage temperature: -20~70 (°C)

dimensional parameters

LCD screen dimensions: 50.00 ± 0.2 (W) x 69.20 ± 0.2 (H) x 2.3 ± 0.1 (D) (mm)
(excluding wiring and adhesive backing)

Touch screen dimensions: 49.60 ± 0.2 (W) x 68.80 ± 0.2 (H) x 1.20 (D) (mm)
(excluding wiring and adhesive backing)

Module dimensions: with touch screen: 50.00 (W) x 86.00 (H) x 5.60 (D) (mm)

No touch screen: 50.00 (W) x 86.00 (H) x 4.40 (D) (mm)

Battery charging parameters

Charging voltage: Range: 4.2~6.5 (V)

Typical value: 5V

Charging current: maximum value: 500mA

Module actual value: 290mA

Charging saturation voltage: 4.24V

Charging temperature: Module actual maximum value: 62 °C

Rechargeable battery specification: 3.7V lithium polymer battery

Electrical parameters:

Working voltage: 5.0V

Backlight current: 75mA

Backlight brightness (actual value) with touch screen: 298 cd/m²

No touch screen: 368 cd/m²

Total current ESP32 reset: 40mA

Only the display screen works: 130mA

Display screen, speaker, and battery charging all work: 480mA

Power consumption: 0.65W (only the display screen works)

Supporting speaker power (maximum): 1.5W (8 Ω) or 2W (4 Ω)

Package Included:

Style 1:

1X development board+1X DuPont Line+1X power cord

Style 2:

1X development board+1X DuPont Line+1X power cord+1X Touchscreen Pen

Electrical Parameters

Operating Voltage: 5.0V

Backlight Current: 75mA

Backlight Brightness (Actual Value): With

Touchscreen: 298 cd/m²

Without Touchscreen: 368 cd/m²

Total Current: ESP32 Reset: 40mA

Display Only: 130mA

Display, Speaker, and Battery Charging All

Working: 480mA

Power Consumption: 0.65W (Display Only)

Supported Speaker Power (Max): 1.5W (8Ω) or
2W (4Ω)

Touchscreen Specifications

Screen Size: 2.8 inch

Screen Type: TN TFT

Resolution: 240xRGBx320 pixels

Effective Display Area: 43.20(W)x-
57.60(H) (mm)

Number of Colors: Maximum: 262K
(RGB666)

Common: 65K (RGB565)

Driver IC: ST7789P3

Display Interface: 4-Line SPI (con-
nected to ESP32)

Pixel Size: 0.153(H)x0.153(mm)

Viewing Angle: 12 O'CLOCK

Backlight Brightness (Typical): 260
cd/m2

Backlight Type: White LED*4

Operating Temperature: -10~50 (°C)

Storage Temperature: -20~60 (°C)



Touchscreen Specifications

Effective touch area size: 2.8 inch

Touchscreen type:

Resistive touchscreen

Driver IC: XPT2046

Effective touch area:

44.20(W) x 58.60(H) (mm)

Viewable window size:

45.20(W) x 59.45(H) (mm)

Communication interface:

SPI

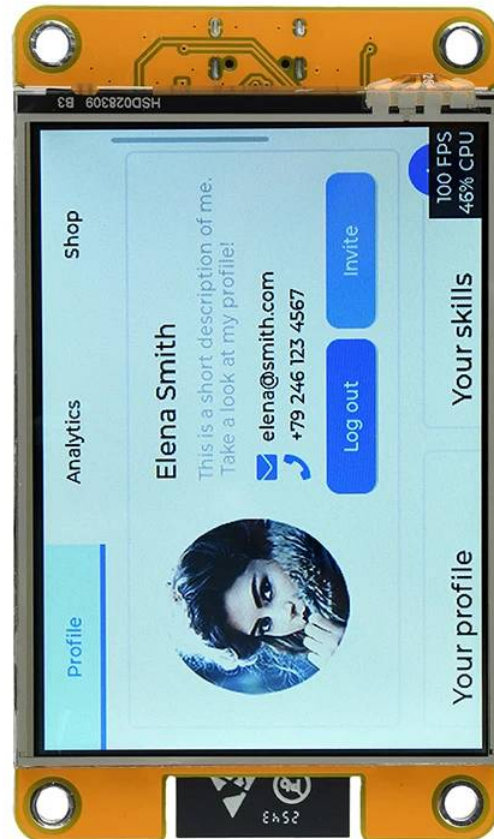
Structure material: ITO film +
ITO glass

Operating temperature:

-10~60(°C)

Storage temperature:

-20~70(°C)



ESP32-32E Main Control Parameters

Module: ESP32-32E

CPU: ESP32-D0WD-V3, Xtensa dual-core 32-bit LX6 microprocessor

Frequency: 240MHz (maximum)

Storage: 448KB ROM + 520KB SRAM + 16KB RTC SRAM + 4MB external
QSPI Flash

WIFI: 2.4GHz, 802.11b/g/n mode

BT: BT V4.2 BR/EDR and BT LE standard

Operating Voltage: 3.0~3.6V





WiFi and BT development board

**Features built-in battery charging management circuitry to
ensure safe battery charging and discharging**



Abundant Interfaces

It features a rich array of interfaces, facilitating connection to various peripherals (SPI, UART, etc.)



240*320 Smart Display Screen (Color)

2.8-inch color screen, 240x320 resolution, supports up to 262K colors (RGB666), displaying rich colors



ESP Development Board

Provides underlying driver technology support and online updates to the WIKI documentation



Single Type-C Interface

Standard Type-C interface for easy program download and power supply



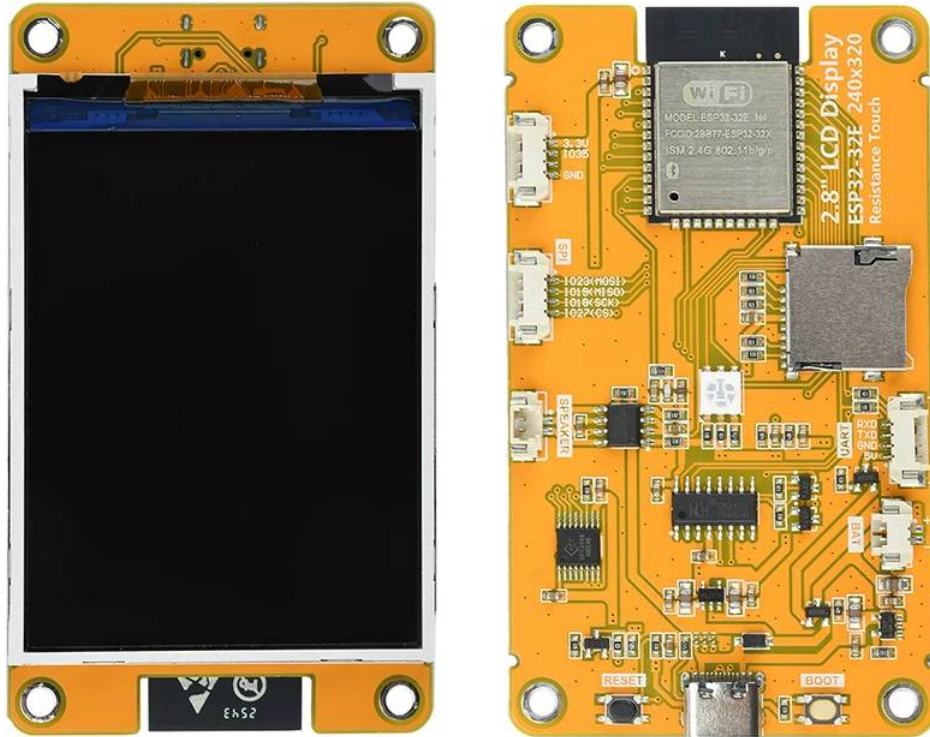
2.8-inch ESP32

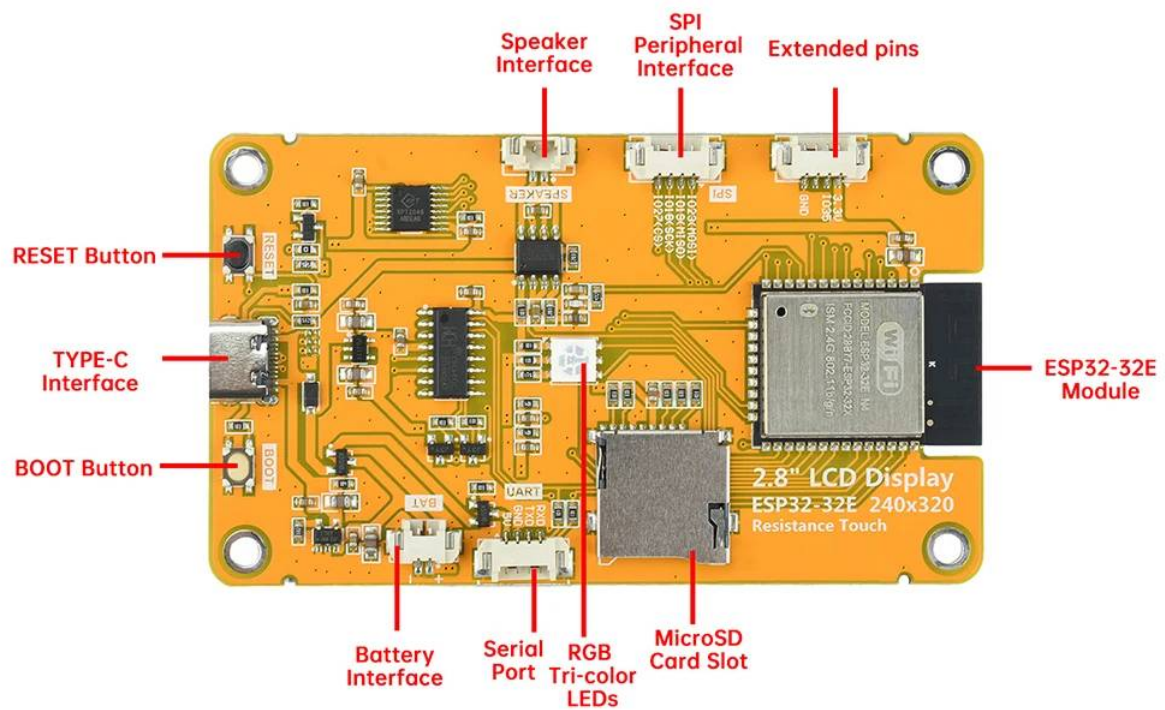
Features built-in RGB tri-color indicator lights, providing a wealth of status indications



ST7789 Driver

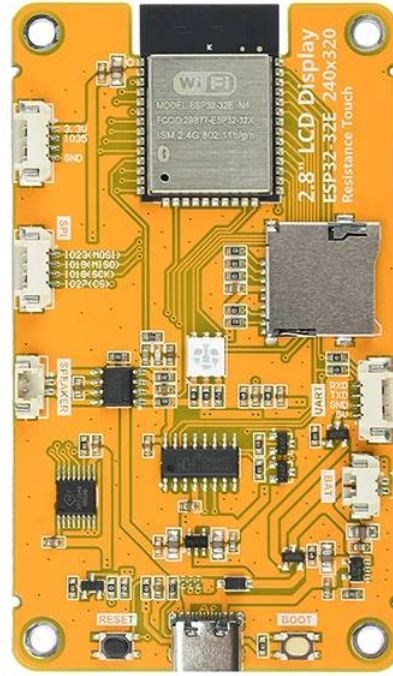
Comes with ESP32-32E module, facilitating development and providing ample development resources



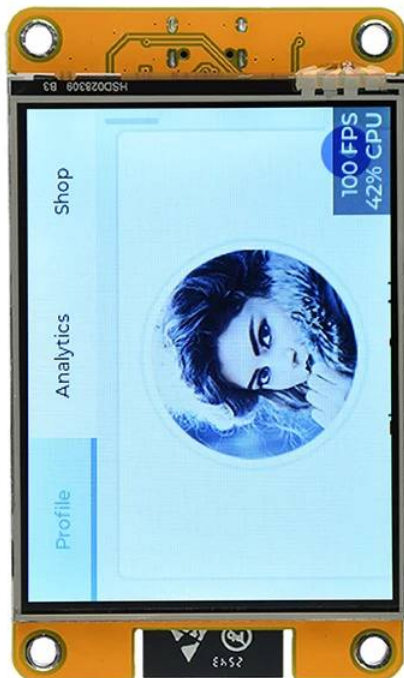




E32N28T-1-TopView

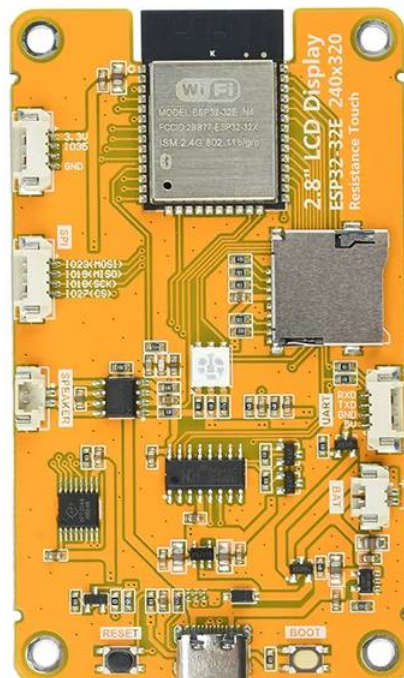


E32N28T-1-BottomView



Driver IC: ST7789

E32N28T-1-TopView



E32N28T-1-BottomView





E32N28T ST7789

No touchscreen, no casing



E32R28T ST7789

With touch screen, no outer casing



E32N28T ST7789

Without touchscreen, with casing



E32R28T ST7789

With touchscreen, with casing

